Fokker-Planck Prime Expansion

1. Graphical & Conceptual Description

This graph visualizes prime number expansion as a diffusion process from a central point.

- **Central Node (C_0):** The origin of the expansion (representing 1, the unit).
- **Branches (B_1 to B_{12}):** 12 'spokes' radiating from the center, like a clock face. These are the primary paths of expansion.
- **Nodes (N_1 to N_{24}):** 24 nodes positioned along each branch, representing 'steps' in the expansion. A node's full address is `(Branch, Node)`, e.g., `(B_7, N_15)`.

The Spiral Algorithm

The first 288 primes (p_1 to p_{288}) are mapped to the 288 nodes in an outward spiral:

- The 1st prime ($p_1=2$) is mapped to `(B_1, N_1)`.
- The 2nd prime ($p_2=3$) is mapped to `(B_2, N_1)`.
- The 12th prime ($p_{12}=37$) is mapped to `(B_12, N_1)`.
- This completes the first 'ring'. The 13th prime ($p_{13}=41$) starts the next ring at `(B_1, N_2)`.
- This continues until the 288th prime ($p_{288}=1861$) is mapped to the final node, `(B_12, N_24)`.

The Modulo 9 Sieve

The `p mod 9` column acts as a 'sieve' or 'filter'. All prime numbers greater than 3 must be congruent to **1, 2, 4, 5, 7, or 8 (mod 9)**.

- **Forbidden Classes:** `p mod 9 = 0, 3, or 6`. Any number in these classes (except 3 itself) is divisible by 3 and thus not prime.
- **Visualizing:** If you 'colored' each node by its `p mod 9` class, you would see the '3', '6', and '9' (or 0) channels are almost entirely empty, visually proving the sieve.

2. Prime Expansion Table (288 Nodes)

Node ID	Prime (p)	Graph Coordinates (Branch, Node)	`p mod 9` (Sieve Class)
1	**2**	`(B_1, N_1)`	2
2	**3**	`(B_2, N_1)`	3
3	**5**	`(B_3, N_1)`	5
4	**7**	`(B_4, N_1)`	7

5	**11**	(B_5, N_1)	2
6	**13**	(B_6, N_1)	4
7	**17**	(B_7, N_1)	8
8	**19**	(B_8, N_1)	1
9	**23**	(B_9, N_1)	5
10	**29**	(B_10, N_1)	2
11	**31**	(B_11, N_1)	4
12	**37**	(B_12, N_1)	1
13	**41**	(B_1, N_2)	5
14	**43**	(B_2, N_2)	7
15	**47**	(B_3, N_2)	2
16	**53**	(B_4, N_2)	8
17	**59**	(B_5, N_2)	5
18	**61**	(B_6, N_2)	7
19	**67**	(B_7, N_2)	4
20	**71**	(B_8, N_2)	8
21	**73**	(B_9, N_2)	1
22	**79**	(B_10, N_2)	7
23	**83**	(B_11, N_2)	2
24	**89**	(B_12, N_2)	8
25	**97**	(B_1, N_3)	7
26	**101**	(B_2, N_3)	2

27	**103**	(B_3, N_3)	4
28	**107**	(B_4, N_3)	8
29	**109**	(B_5, N_3)	1
30	**113**	(B_6, N_3)	5
31	**127**	(B_7, N_3)	1
32	**131**	(B_8, N_3)	5
33	**137**	(B_9, N_3)	2
34	**139**	(B_10, N_3)	4
35	**149**	(B_11, N_3)	5
36	**151**	(B_12, N_3)	7
37	**157**	(B_1, N_4)	4
38	**163**	(B_2, N_4)	1
39	**167**	(B_3, N_4)	5
40	**173**	(B_4, N_4)	2
41	**179**	(B_5, N_4)	8
42	**181**	(B_6, N_4)	1
43	**191**	(B_7, N_4)	2
44	**193**	(B_8, N_4)	4
45	**197**	(B_9, N_4)	8
46	**199**	(B_10, N_4)	1
47	**211**	(B_11, N_4)	4
48	**223**	(B_12, N_4)	7

49	**227**	(B_1, N_5)	2
50	**229**	(B_2, N_5)	4
51	**233**	(B_3, N_5)	8
52	**239**	(B_4, N_5)	5
53	**241**	(B_5, N_5)	7
54	**251**	(B_6, N_5)	8
55	**257**	(B_7, N_5)	5
56	**263**	(B_8, N_5)	2
57	**269**	(B_9, N_5)	8
58	**271**	(B_10, N_5)	1
59	**277**	(B_11, N_5)	7
60	**281**	(B_12, N_5)	2
61	**283**	(B_1, N_6)	4
62	**293**	(B_2, N_6)	5
63	**307**	(B_3, N_6)	1
64	**311**	(B_4, N_6)	5
65	**313**	(B_5, N_6)	7
66	**317**	(B_6, N_6)	2
67	**331**	(B_7, N_6)	7
68	**337**	(B_8, N_6)	4
69	**347**	(B_9, N_6)	5
70	**349**	(B_10, N_6)	7

71	**353**	(B_11, N_6)	2
72	**359**	(B_12, N_6)	8
73	**367**	(B_1, N_7)	7
74	**373**	(B_2, N_7)	4
75	**379**	(B_3, N_7)	1
76	**383**	(B_4, N_7)	5
77	**389**	(B_5, N_7)	2
78	**397**	(B_6, N_7)	1
79	**401**	(B_7, N_7)	5
80	**409**	(B_8, N_7)	4
81	**419**	(B_9, N_7)	5
82	**421**	(B_10, N_7)	7
83	**431**	(B_11, N_7)	8
84	**433**	(B_12, N_7)	1
85	**439**	(B_1, N_8)	7
86	**443**	(B_2, N_8)	2
87	**449**	(B_3, N_8)	8
88	**457**	(B_4, N_8)	7
89	**461**	(B_5, N_8)	2
90	**463**	(B_6, N_8)	4
91	**467**	(B_7, N_8)	8
92	**479**	(B_8, N_8)	2

93	**487**	(B_9, N_8)	1
94	**491**	(B_10, N_8)	5
95	**499**	(B_11, N_8)	4
96	**503**	(B_12, N_8)	8
97	**509**	(B_1, N_9)	5
98	**521**	(B_2, N_9)	8
99	**523**	(B_3, N_9)	1
100	**541**	(B_4, N_9)	1
101	**547**	(B_5, N_9)	7
102	**557**	(B_6, N_9)	2
103	**563**	(B_7, N_9)	5
104	**569**	(B_8, N_9)	2
105	**571**	(B_9, N_9)	4
106	**577**	(B_10, N_9)	1
107	**587**	(B_11, N_9)	2
108	**593**	(B_12, N_9)	8
109	**599**	(B_1, N_10)	5
110	**601**	(B_2, N_10)	7
111	**607**	(B_3, N_10)	4
112	**613**	(B_4, N_10)	1
113	**617**	(B_5, N_10)	5
114	**619**	(B_6, N_10)	7

115	**631**	(B_7, N_10)	1
116	**641**	(B_8, N_10)	2
117	**643**	(B_9, N_10)	4
118	**647**	(B_10, N_10)	8
119	**653**	(B_11, N_10)	5
120	**659**	(B_12, N_10)	2
121	**661**	(B_1, N_11)	4
122	**673**	(B_2, N_11)	7
123	**677**	(B_3, N_11)	2
124	**683**	(B_4, N_11)	8
125	**691**	(B_5, N_11)	7
126	**701**	(B_6, N_11)	8
127	**709**	(B_7, N_11)	7
128	**719**	(B_8, N_11)	8
129	**727**	(B_9, N_11)	7
130	**733**	(B_10, N_11)	4
131	**739**	(B_11, N_11)	1
132	**743**	(B_12, N_11)	5
133	**751**	(B_1, N_12)	4
134	**757**	(B_2, N_12)	1
135	**761**	(B_3, N_12)	5
136	**769**	(B_4, N_12)	4

137	**773**	(B_5, N_12)	8
138	**787**	(B_6, N_12)	4
139	**797**	(B_7, N_12)	5
140	**809**	(B_8, N_12)	8
141	**811**	(B_9, N_12)	1
142	**821**	(B_10, N_12)	2
143	**823**	(B_11, N_12)	4
144	**827**	(B_12, N_12)	8
145	**829**	(B_1, N_13)	1
146	**839**	(B_2, N_13)	2
147	**853**	(B_3, N_13)	7
148	**857**	(B_4, N_13)	2
149	**859**	(B_5, N_13)	4
150	**863**	(B_6, N_13)	8
151	**877**	(B_7, N_13)	4
152	**881**	(B_8, N_13)	8
153	**883**	(B_9, N_13)	1
154	**887**	(B_10, N_13)	5
155	**907**	(B_11, N_13)	7
156	**911**	(B_12, N_13)	2
157	**919**	(B_1, N_14)	1
158	**929**	(B_2, N_14)	2

159	**937**	(B_3, N_14)	1
160	**941**	(B_4, N_14)	5
161	**947**	(B_5, N_14)	2
162	**953**	(B_6, N_14)	8
163	**967**	(B_7, N_14)	4
164	**971**	(B_8, N_14)	8
165	**977**	(B_9, N_14)	5
166	**983**	(B_10, N_14)	2
167	**991**	(B_11, N_14)	1
168	**997**	(B_12, N_14)	7
169	**1009**	(B_1, N_15)	1
170	**1013**	(B_2, N_15)	5
171	**1019**	(B_3, N_15)	2
172	**1021**	(B_4, N_15)	4
173	**1031**	(B_5, N_15)	5
174	**1033**	(B_6, N_15)	7
175	**1039**	(B_7, N_15)	4
176	**1049**	(B_8, N_15)	5
177	**1051**	(B_9, N_15)	7
178	**1061**	(B_10, N_15)	8
179	**1063**	(B_11, N_15)	1
180	**1069**	(B_12, N_15)	7

181	**1087**	(B_1, N_16)	7
182	**1091**	(B_2, N_16)	2
183	**1093**	(B_3, N_16)	4
184	**1097**	(B_4, N_16)	8
185	**1103**	(B_5, N_16)	5
186	**1109**	(B_6, N_16)	2
187	**1117**	(B_7, N_16)	1
188	**1123**	(B_8, N_16)	7
189	**1129**	(B_9, N_16)	4
190	**1151**	(B_10, N_16)	8
191	**1153**	(B_11, N_16)	1
192	**1163**	(B_12, N_16)	2
193	**1171**	(B_1, N_17)	1
194	**1181**	(B_2, N_17)	2
195	**1187**	(B_3, N_17)	8
196	**1193**	(B_4, N_17)	5
197	**1201**	(B_5, N_17)	4
198	**1213**	(B_6, N_17)	7
199	**1217**	(B_7, N_17)	2
200	**1223**	(B_8, N_17)	8
201	**1229**	(B_9, N_17)	5
202	**1231**	(B_10, N_17)	7

203	**1237**	(B_11, N_17)	4
204	**1249**	(B_12, N_17)	7
205	**1259**	(B_1, N_18)	8
206	**1277**	(B_2, N_18)	8
207	**1279**	(B_3, N_18)	1
208	**1283**	(B_4, N_18)	5
209	**1289**	(B_5, N_18)	2
210	**1291**	(B_6, N_18)	4
211	**1297**	(B_7, N_18)	1
212	**1301**	(B_8, N_18)	5
213	**1303**	(B_9, N_18)	7
214	**1307**	(B_10, N_18)	2
215	**1319**	(B_11, N_18)	5
216	**1321**	(B_12, N_18)	7
217	**1327**	(B_1, N_19)	4
218	**1361**	(B_2, N_19)	2
219	**1367**	(B_3, N_19)	8
220	**1373**	(B_4, N_19)	5
221	**1381**	(B_5, N_19)	4
222	**1399**	(B_6, N_19)	4
223	**1409**	(B_7, N_19)	5
224	**1423**	(B_8, N_19)	1

225	**1427**	(B_9, N_19)	5
226	**1429**	(B_10, N_19)	7
227	**1433**	(B_11, N_19)	2
228	**1439**	(B_12, N_19)	8
229	**1447**	(B_1, N_20)	7
230	**1451**	(B_2, N_20)	2
231	**1453**	(B_3, N_20)	4
232	**1459**	(B_4, N_20)	1
233	**1471**	(B_5, N_20)	4
234	**1481**	(B_6, N_20)	5
235	**1483**	(B_7, N_20)	7
236	**1487**	(B_8, N_20)	2
237	**1489**	(B_9, N_20)	4
238	**1493**	(B_10, N_20)	8
239	**1499**	(B_11, N_20)	5
240	**1511**	(B_12, N_20)	8
241	**1523**	(B_1, N_21)	2
242	**1531**	(B_2, N_21)	1
243	**1543**	(B_3, N_21)	4
244	**1549**	(B_4, N_21)	1
245	**1553**	(B_5, N_21)	5
246	**1559**	(B_6, N_21)	2

247	**1567**	(B_7, N_21)	1
248	**1571**	(B_8, N_21)	5
249	**1579**	(B_9, N_21)	4
250	**1583**	(B_10, N_21)	8
251	**1597**	(B_11, N_21)	4
252	**1601**	(B_12, N_21)	8
253	**1607**	(B_1, N_22)	5
254	**1609**	(B_2, N_22)	7
255	**1613**	(B_3, N_22)	2
256	**1619**	(B_4, N_22)	8
257	**1621**	(B_5, N_22)	1
258	**1627**	(B_6, N_22)	7
259	**1637**	(B_7, N_22)	8
260	**1657**	(B_8, N_22)	1
261	**1663**	(B_9, N_22)	7
262	**1667**	(B_10, N_22)	2
263	**1669**	(B_11, N_22)	4
264	**1693**	(B_12, N_22)	1
265	**1697**	(B_1, N_23)	5
266	**1699**	(B_2, N_23)	7
267	**1709**	(B_3, N_23)	8
268	**1721**	(B_4, N_23)	2

269	**1723**	(B_5, N_23)	4
270	**1733**	(B_6, N_23)	5
271	**1741**	(B_7, N_23)	4
272	**1747**	(B_8, N_23)	1
273	**1753**	(B_9, N_23)	7
274	**1759**	(B_10, N_23)	4
275	**1777**	(B_11, N_23)	4
276	**1783**	(B_12, N_23)	1
277	**1787**	(B_1, N_24)	5
278	**1789**	(B_2, N_24)	7
279	**1801**	(B_3, N_24)	1
280	**1811**	(B_4, N_24)	2
281	**1823**	(B_5, N_24)	5